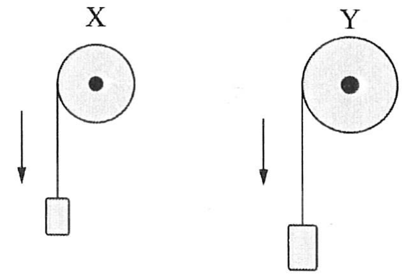
AP Physics 1 Quiz 5.1 Rotational Motion Name: Sunny

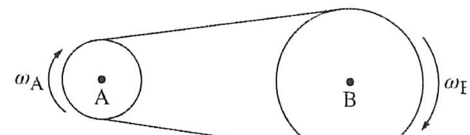
A Side

B 1. Pulleys X and Y are each attached to a block by a string that wraps around the pulley. Both blocks are released from rest and have the same linear acceleration a. As the blocks fall, the pulleys rotate about their centers. Pulley Y has a larger radius than Pulley X. How does the final angular velocity ω_Y of Pulley Y compare to the final angular velocity ω_X of Pulley X?



(A) $\omega_Y > \omega_X$
(B) $\omega_Y < \omega_X$
(C) $\omega_Y = \omega_X$
(D) It cannot be determined without knowing the relative masses of each block.

A 2. Wheels A and B are connected by a moving belt and are both free to rotate about their centers, as shown. The belt does not slip on the wheels. The radius of Wheel B is three times larger than the radius of Wheel A. Wheel A has constant angular speed ω_A and Wheel B has constant angular speed ω_B. Which of the following correctly relates ω_A



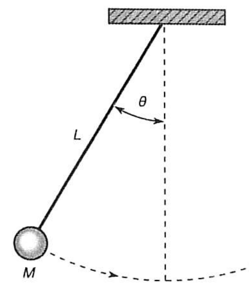
(A) $\omega_A = \frac{\omega_B}{3}$
(B) $\omega_A = \omega_B$
(C) $\omega_A = 3\omega_B$
(D) $\omega_A = 6\omega_B$

$V_A = V_B$

$r_A = \frac{1}{3} r_B$

$r_A \omega_A = r_B \omega_B$

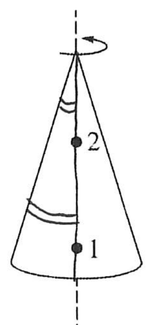
C 3. A pendulum is formed from a mass M connected to a string of length L. The pendulum is initially raised to form an angle θ (measured in radians) and released. The time it takes the pendulum to swing to the opposite side and return to its initial position is measured to be T (measured in seconds). Which of the following is a correct expression for the average linear speed of the pendulum?



(A) $\frac{\theta}{T}$
(B) $\frac{\theta L}{T}$
(C) $\frac{2\theta L}{T}$
(D) $\frac{4\theta L}{T}$

$\frac{\Delta x}{\Delta t}$ arc length $\theta L \times 2$

D 4. A solid, uniform cone is spinning with constant angular speed about a vertical axis through its center of mass, as shown. A small insect is initially standing on the surface of the cone at Point 1. At a later time, the insect moved to Point 2. Which of the following statements about the motion of the insect is true?



(A) The angular speed of the insect is greater at Point 1 than at Point 2.
(B) The angular speed of the insect is greater at Point 2 than at Point 1.
(C) The linear speed of the insect is greater at Point 1 than at Point 2.
(D) The linear speed of the insect is greater at Point 2 than at Point 1.

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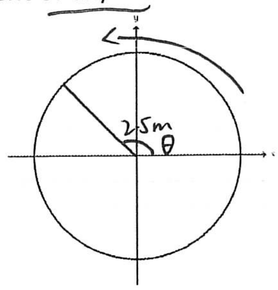
B Side

1. A disk has a radius of 2.5 m and undergoes an angular displacement of 3π/4 radians.

r θ

a. Sketch this angular displacement



b. Convert this measurement to degrees.

$\frac{3\pi}{4} = \frac{180° \times 3}{4} = 135°$

c. Determine the arc length formed.

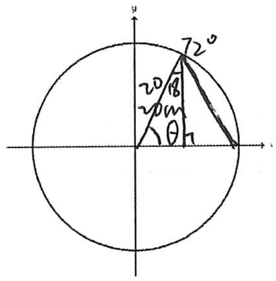
arc length $= \theta \cdot r = \frac{3\pi}{4} \times 2.5m$

$= \frac{7.5\pi}{4} = \frac{15\pi}{8} = 1.875\pi = 5.89$ m.

2. A woman walks in a circular path of radius 20 m. The woman undergoes an angular displacement of 2.2 rotations in the counterclockwise direction, which takes her 20 seconds to complete.

positive

a. Sketch this angular displacement



b. Convert her angular displacement to radians.

2.2 rotation $= 792° = \frac{22\pi}{5}$.

c. Calculate the woman's linear displacement.

$360° \times (2.2-2) = 72°$.

$\cancel{\frac{22\pi}{5} \times 20 = 88\pi}$.

d. Calculate the woman's angular velocity (in rad/s)

$\cancel{\omega = \frac{v}{r} = \frac{0.11}{20m} = 0.0055 \text{ rad/s}}$ ~~2.2 ÷ 20 = 0.11 rotations/s~~

$\omega = \frac{\Delta\theta}{t} = \frac{\frac{22\pi}{5}}{20} = 0.22\pi \approx 0.691$ rad/s.

e. Calculate the woman's linear velocity (in m/s)

$v = r\omega =$ ~~[illegible]~~ $20 \times 0.22\pi = 4.4\pi$

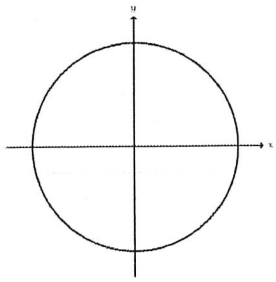
$= 13.82$ m/s

3. A record has a small red dot on its edge. You measure that the dot travels a distance of 1.3 m (clockwise) when the record makes one full rotation, which takes 1.2 seconds.

2π

a. Determine the radius of the record

$r = \frac{1.3}{2\pi} = 0.207$ m.



b. Determine the angular velocity of the record (in deg/s)

$\omega = \frac{\Delta\theta}{t} = \frac{360°}{1.2} = 300$ deg/s